Japan’s semiconductors are more than manufacturing

SEMICONDUCTOR DESIGN | Japanese strengths | As of September 24, 2026

Focus on design, IP and custom SoCs

In advanced logic semiconductors, US firms are prominent in design and Taiwanese firms in manufacturing. When assessing Japanese companies, look beyond manufacturing share to firms creating added value through design.

Creating chips without owning factories

Fabless firms own no manufacturing plants. They design circuits and SoCs to customer specifications and outsource manufacturing, operating without carrying huge production facilities themselves.

Four areas linked to demand for dedicated chips

Custom SoC / ASIC: chips dedicated to customers’ applications.
LSI design: planning and circuit design for large-scale integrated circuits.
GPU / NPU and other IP: reusable design assets for graphics processing and AI computation.
High-speed interfaces: technologies for rapid data transmission between chips and devices.

AI, data centers, automotive applications, robots and high-speed communications are expected to generate demand for application-optimized semiconductors.

Five stocks involved in semiconductor design

JAPANESE LISTED FIRMS | Design, fabless and design IP | September 24, 2026

Socionext / 6526 | Custom SoCs and ASICs

Supports customers from product planning and specification through design, production and quality control. Designs advanced SoCs for data centers, networks and automotive applications.

MegaChips / 6875 | ASIC, ASSP and system LSI

A fabless model supporting system design using CPUs and various IP, low-power design and design for testability. Broad applications include communications, factory automation and image processing.

THine Electronics / 6769 | Mixed-signal and high-speed interfaces

Fabless firm planning and designing LSIs combining analog and digital technology. Focuses on high-speed transmission and power control, with strengths in peripheral technologies moving data rapidly.

Digital Media Professionals / 3652 | GPU/NPU IP and edge-AI SoCs

Known as DMP. Offers GPU/AI IP licensing and semiconductor products. Design assets include NPUs, 2D/3D GPUs and stereo vision, linking it to image recognition and robots.

Axell / 6730 | Application-specific LSI and image/audio processing

LSI design cultivated for amusement applications is its core. The focus is whether it can expand image/audio processing and other technologies into additional applications and markets.

Four ways to assess design themes

TECHNOLOGY AND EARNINGS STRUCTURE | As of September 24, 2026

1. Can the company participate upstream in design?

Check involvement in customer product planning, overall architecture and specifications, beyond circuit layout and wiring. Upstream proposal capabilities create added value.

2. Does it own reusable IP?

GPU, NPU and high-speed interface design IP can be deployed across products. Examine how license fees and royalties contribute to earnings.

3. Does it connect with growth applications?

Generative AI, data centers, networks, automotive advanced driver-assistance systems and robots. Verify contributions to orders and sales, beyond mere association.

4. Understand dependence on external manufacturing

Fabless firms can limit capital spending but depend on external semiconductor foundries. Access to advanced processes, production capacity and manufacturing costs pose risks.

Semiconductor stocks play different roles

Equipment, materials, design/IP and IDMs integrating design and manufacturing have different earnings structures. These five companies are organized around design itself and design assets.

Use global manufacturing capacity through design

Japan’s semiconductor story extends beyond recovering manufacturing share. Firms leveraging global production capacity through custom SoCs and design IP also merit attention.